



COMPUTATIONAL ORDINAL SOCIOMETRY AND SOCIO-THERMODYNAMIC DIAGNOSTICS (NEXORD PLATFORM)

Scientific Manifesto, Theoretical-Methodological Framework, and Institutional Governance Guidelines

«Hacia una Física de lo Grupal_JMC»

Canonical Certified Version v8.0 (2026)

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Concept DOI (Zenodo / CERN): [10.5281/zenodo.18926396](https://doi.org/10.5281/zenodo.18926396)

Homologated Edition DOI: [10.5281/zenodo.18941691](https://doi.org/10.5281/zenodo.18941691)

Official Web Platform & Interactive Ecosystem: <https://josemancor.github.io/nex-ord-web/>

Official Repository & Source Code: <https://github.com/josemancor/nex-ord-web>

Master Brand / Ecosystem: VISORD 3D/4D (Visual Ordinal Sociometry & Socio-Thermodynamic Diagnostics)

Platform Supported Languages: Spanish (ES), English (EN), French (FR), German (DE), Italian (IT), Portuguese (PT), Russian (RU), Chinese (ZH).

Executive Abstract

Computational Ordinal Sociometry (NEXORD) is, in essence, an updated extension of the foundational insights of Jacob Levy Moreno, creator of Sociometry, and Kurt Lewin, pioneer of field theory—two seminal figures in Social Psychology. Historically, group dynamics research has been constrained by the methodological and technical limitations of its founding era. This Manifesto is grounded in:

- 1) The conception of the ‘relational tie’, born of human interaction, as the core constituting the existential reality of groups. ‘Giving and Receiving’ are indissociable force pairs that sustain the necessary dialectical exchange of affect, ideas, intentions, resources, and collective actions toward shared goals. Personal and group health, wellbeing, success, and viability fundamentally depend on the continuous homeostatic balance established across interpersonal exchanges of acceptance and conflict.
- 2) The refined quantification of preferential relational order through the inverse rank function, together with contemporary theoretical, methodological, and technological developments, defines the genesis of NEXORD as an overarching sociometric platform operationalizing Moreno’s and Lewin’s core intuitions.
- 3) The continuous metric emerging from this quantification is formally articulated, through matrix and tensorial algebra, within the Moreno-rooted relational tetragram (E, R, pE, pR) across $3^4 = 81$ dyadic configurations operationalized via the SMib/SMIa matrix duality, translating subjective micro-interaction into computable topology.
- 4) The articulation of the 4 elemental densities and their analytical extension to 18 derived relational densities (18D space) from the Q81 lattice, quantifying the mass and flux of exchanges. This continuous gradient models the collective as a socio-thermodynamic system governed by equilibrium regimes and relational friction (Φ), where dialectical friction acts as a necessary condition for systemic health and adaptability against groupthink entropy.
- 5) Simultaneous projection across dual factorial spaces ($CA \parallel PCA$), wherein actors emerge not as static scalar points, but as dyadic kinematic dipoles ($DA \rightarrow RECEIVE$) whose centroid analytically equilibrates emission and reception. This formulation reveals the gravitational dominance of the primary relational axis ($PC_1 \gg PC_2$) and enables scale-invariant geometric comparison of group topologies across Grassmann manifolds $Gr(k, N)$.

6) The versatility of the multidimensional design ($G \times T \times C$), enabling integrated transversal, longitudinal, and criterial comparisons (intra- and inter-group). This architecture is enhanced by the interactive VISORD visualization modules, facilitating the monitoring of sociometric structures and processes emerging from the analyses.

7) NEXORD's effective ethical commitment to a strictly constructive and non-punitive sociometric praxis, ensuring absolute privacy protection by design (*Privacy by Design*). This framework mandates institutional oversight and the explicit deontological commitment of the accredited Principal Investigator (PI) as sole interlocutor, under Open Science standards.

Keywords: NEXORD Platform, VISORD, Computational Ordinal Sociometry, Preventive Sociometry, Breve Format, Inverse Rank Function, Relational Densities (18D), Metaperception, Dual Factorial Planes, Grassmann Manifolds, Groupthink, Ostracism, Ethics by Design, Open Science.

1. Epistemological Foundations

1.1. Overcoming the Classical Limit & the Ontology of the Relational Tie

NEXORD's objective is the methodological updating of social psychology traditions: translating into computational formalisms the foundational insights that neither Jacob L. Moreno nor Kurt Lewin could operationalize with the technical tools of their era. Moreno introduced the concept of the social atom, but manual computation restricted sociometry to fixed binary quotas (typically three choices and three rejections). Lewin formulated psychological field theory and social topology, but lacked continuous mathematical operators to quantify relational distances without distorting the psychological nature of interpersonal ties. NEXORD places active relational interaction at the ontological core of group reality: the collective is not an aggregate of isolated nodes, but a dynamic field of forces in continuous emergence.

1.2. Inverse Rank Quantification

The mathematical foundation enabling the transition to continuous metrics is the quantification of hierarchical order through the inverse rank function:

$$W_{ij} = \frac{1}{r_{ij}^\gamma} \quad (\gamma > 0)$$

Formula 1: Relational Decay

where r_{ij} in $\{1, 2, \dots, N\}$ represents the strict ordinal rank assigned by sender i to receiver j , and $\gamma > 0$ governs the field's openness and relational decay rate. This formulation translates qualitative ordering into continuous relational values, allowing dyadic ties to be operated as measurable forces projectable into multidimensional spaces. This formal principle aligns with statistical frameworks for rank-order relational data (Krivitsky & Butts, 2017) and psychometric findings on measurement degradation in truncated discrete nominations (Terry, 2000).

1.3. The Fundamental Axiom: GIVING and RECEIVING as an Indissociable Circuit

Emitting choices (GIVING) and receiving collective evaluations (RECEIVING) do not represent independent variables or isolated traits: they are immanent and indissociable moments of a single continuous relational circuit. The functional health or strain of a group does not reside within isolated members, but in the fluidity, symmetry, and friction of their mutual exchanges. An actor may emit positive choices while facing systematic collective exclusion; detaching these two polar moments produces incomplete diagnostics blind to latent structural isolation.

2. Capture Protocol: BREVE Format, Matrix Duality & Diagonal Closure

2.1. Forced Drag without Ties & Matrix Duality: A(i, rank) vs. B(i, j)

In the BREVE Format, participants view a randomized peer list to eliminate order biases. Through a forced drag-and-drop task, they rank all group members between maximum preference and rejection poles. The zero-ties rule ($r_{ij} \neq r_{ik}$) eliminates acquiescence bias and ensures complete ranking. This capture is mathematically formalized into two complementary matrices:

- Matrix A(i, rank) BREVE:** Represents direct ordinal sequences emitted by each participant. Rows represent senders i , columns represent ordinal ranks $k \in \{1st, \dots, Nth\}$, and cell contents record the peer placed at that rank: $A(i, k) = j$.
- Matrix B(i, j) RANK DISTRIBUTION:** Square sociomatrix ($N \times N$) where rows represent senders i , columns represent receivers j , and cell values record the assigned numerical rank: $B(i, j) = k \Leftrightarrow A(i, k) = j$.

Table 1A: Factual Formalization of Group A (G_1 , $N = 6$, Bipolar 3 + 3) — BREVE Format, Classical Moreno, and Metaperceptions

Node ID	1st	2nd	3rd	4th	5th	6th	Choices (E)	Rejections (R)	Expected Choices (pE)	Expected Rejections (pR)	αE (Closest)	ωE (Distant)
1A	2A	1A	3A	5A	4A	6A	2A, 3A	4A, 6A	2A, 3A	4A, 6A	2A	2A → 1A
2A	1A	2A	3A	4A	6A	5A	1A, 3A	5A, 6A	1A, 3A	5A, 6A	1A	1A → 2A
3A	1A	2A	3A	6A	5A	4A	1A, 2A	4A, 5A	1A, 2A	4A, 5A	1A	1A → 2A
4A	5A	4A	6A	2A	1A	3A	5A, 6A	1A, 3A	5A, 6A	1A, 3A	5A	5A → 4A
5A	4A	5A	6A	1A	3A	2A	4A, 6A	2A, 3A	4A, 6A	2A, 3A	4A	4A → 5A
6A	4A	5A	6A	3A	2A	1A	4A, 5A	1A, 2A	4A, 5A	1A, 2A	4A	4A → 5A

Table 1B: Instrumental SMIb Sociomatrix of Group A (G_1 , $N = 6$) — Dyadic Tetragram ($A_1A_2A_3A_4 = pDA \cdot DA \cdot RECIBE \cdot pRECIBE$)

G_1	1A	2A	3A	4A	5A	6A
1A	0AA0	ABBA	ABBA	0ab0	0ab0	0ab0
2A	BAAB	0AA0	ABBA	0ab0	0ab0	0ab0
3A	BAAB	BAAB	0AA0	0ab0	0ab0	0ab0
4A	0ba0	0ba0	0ba0	0AA0	ABBA	ABBA
5A	0ba0	0ba0	0ba0	BAAB	0AA0	ABBA
6A	0ba0	0ba0	0ba0	BAAB	BAAB	0AA0

Table 1C: Integrated SMIa Sociomatrix of Group A (G_1 , $N = 6$) — Canonical Q81 Lattice

G_1	1A	2A	3A	4A	5A	6A	Subgroup Diagnostic
1A	i¿?!	<Ee>	<Ee>	[Rr]	[Rr]	[Rr]	Cohesion {1A, 2A, 3A} / Opposition {4A, 5A, 6A}
2A	<Ee>	i¿?!	<Ee>	[Rr]	[Rr]	[Rr]	Cohesion {1A, 2A, 3A} / Opposition {4A, 5A, 6A}
3A	<Ee>	<Ee>	i¿?!	[Rr]	[Rr]	[Rr]	Cohesion {1A, 2A, 3A} / Opposition {4A, 5A, 6A}
4A	[Rr]	[Rr]	[Rr]	i¿?!	<Ee>	<Ee>	Cohesion {4A, 5A, 6A} / Opposition {1A, 2A, 3A}
5A	[Rr]	[Rr]	[Rr]	<Ee>	i¿?!	<Ee>	Cohesion {4A, 5A, 6A} / Opposition {1A, 2A, 3A}
6A	[Rr]	[Rr]	[Rr]	<Ee>	<Ee>	i¿?!	Cohesion {4A, 5A, 6A} / Opposition {1A, 2A, 3A}

Table 1D: Factual Formalization of Group B (G_2 , $N = 6$, Systematic Ostracism of 6B) — BREVE Format and Inbound Status (SRC)

ID	1st	2nd	3rd	4th	5th	6th (Rejection)	Node 6B in SMIa	Inbound Status (SRC)	Structural Diagnosis
1B	2B	1B	3B	4B	5B	6B	1B → 6B: [Rr]	+0.875	Cohesive Attractor Leader
2B	1B	2B	3B	5B	4B	6B	2B → 6B: [Rr]	+1.250	Primary Attractor Leader
3B	1B	2B	3B	4B	5B	6B	3B → 6B: [Rr]	+0.625	Cohesive Core
4B	3B	4B	1B	2B	5B	6B	4B → 6B: [Rr]	+0.125	Integrated Neutral
5B	2B	3B	5B	1B	4B	6B	5B → 6B: [Rr]	+0.125	Integrated Neutral
6B	1B	2B	6B	3B	4B	5B	6B → Group: <Er]	−1.500	Unanimous Ostracism (Relational Asphyxia)

FIGURE 1: BREVE INPUT ARCHITECTURE AND DERIVED SMib / SMla MATRICES
Factual Formalization of Bipolar Group G1 (N = 6, Two 3+3 Subgroups) · Moreno Bijjective Equivalence and Q81 Relational Lattice

ID	BREVE (Rankings 1st to 6th)						CLASSICAL MORENO				METAPERCEPTIONS	
	1º	2º	3º	4º	5º	6º	E	R	pE	pR	αE	ωE
1A	2A	1A	3A	5A	4A	6A	2A, 3A	4A, 6A	2A, 3A	4A, 6A	2A	2A→1A
2A	1A	2A	3A	4A	6A	5A	1A, 3A	5A, 6A	1A, 3A	5A, 6A	1A	1A→2A
3A	1A	2A	3A	6A	5A	4A	1A, 2A	4A, 5A	1A, 2A	4A, 5A	1A	1A→2A
4A	5A	4A	6A	2A	1A	3A	5A, 6A	1A, 3A	5A, 6A	1A, 3A	5A	5A→4A
5A	4A	5A	6A	1A	3A	2A	4A, 6A	2A, 3A	4A, 6A	2A, 3A	4A	4A→5A
6A	4A	5A	6A	3A	2A	1A	4A, 5A	1A, 2A	4A, 5A	1A, 2A	4A	4A→5A

G1	Instrumental SMib MATRIX					
	1A	2A	3A	4A	5A	6A
1A	0AA0	ABBA	ABBA	0ab0	0ab0	0ab0
2A	BAAB	0AA0	ABBA	0ab0	0ab0	0ab0
3A	BAAB	BAAB	0AA0	0ab0	0ab0	0ab0
4A	0ba0	0ba0	0ba0	0AA0	ABBA	ABBA
5A	0ba0	0ba0	0ba0	BAAB	0AA0	ABBA
6A	0ba0	0ba0	0ba0	BAAB	BAAB	0AA0

G1	Integrated SMla MATRIX					
	1A	2A	3A	4A	5A	6A
1A	i ₂ ?	<Ee>	<Ee>	[Rr]	[Rr]	[Rr]
2A	<Ee>	i ₂ ?	<Ee>	[Rr]	[Rr]	[Rr]
3A	<Ee>	<Ee>	i ₂ ?	[Rr]	[Rr]	[Rr]
4A	[Rr]	[Rr]	[Rr]	i ₂ ?	<Ee>	<Ee>
5A	[Rr]	[Rr]	[Rr]	<Ee>	i ₂ ?	<Ee>
6A	[Rr]	[Rr]	[Rr]	<Ee>	<Ee>	i ₂ ?

METHODOLOGICAL INTERPRETATION NOTE (NEXORD Canonical Rules):
1. Bijjective Equivalence BREVE <-> MORENO: Group A (1A..6A). In BREVE, each row ranks all peers from 1st to 6th; Moreno extracts choices (E), rejections (R), and metaperceptions.
2. SMib Tetragram (A1, A2, A3, A4): p, DA, DA, RECIBE, p, RECIBE (Uppercase = Choice, Lowercase = Rejection, 0 = Neutral).
3. Q81 Lattice: <Ee>=Mutual Choice (Green), [Rr]=Mutual Rejection (Red/Magenta), i₂? = Reflexive Diagonal Closure (Blue).

Figure 1: BREVE Input Architecture and Derived SMib / SMla Matrices (Canonical N = 6 Scenario, Bipolar Group G1). Left Panel: Forced ranking Matrix A(i, rank) 1st to 6th, bijjective Moreno transcription (E, R, pE, pR), and triangulated metaperceptions (αE, ωE). Center Panel: Instrumental SMib sociomatrix of dyadic tetragrams (A1:A2-A3:A4). Right Panel: Integrated SMla sociomatrix in Q81 lattice.

2.2. Reflexive Diagonal Closure & Core Sensors

NEXORD resolves the classical sociometric diagonal omission (traditionally empty, r_{ii} = empty) by requiring participants to self-anchor within the ranking (r_{ii} in {1, ..., N}). This reflexive closure equips the system with three analytical sensors:

- **IDA (Self-Rank Misalignment Index):** Angular difference between self-image (r_{ii}) and received peer consensus (R_{ext_i}):

$$IDA_i = \frac{r_{ii} - \bar{R}_{ext_i}}{N-1} \in [-1.00, +1.00]$$

Formula 2: Self-Image Discrepancy IDA

where IDA < 0 marks status overestimation, IDA > 0 indicates self-deprecation or timidity, and IDA = 0 reflects balanced social calibration. At the collective level, group deviation is computed quadratically:

$$IDA_{\text{grupo}} = \sqrt{\frac{1}{N} \sum_{i=1}^N IDA_i^2}$$

Formula 3: Group Discrepancy IDA_{group}

- **ITA (Self-Rank Tension Index):** Evaluates structural strain arising when multiple members claim rank one (r_{ii} = 1).
- **Cognitive Partitioning P(k):** By positioning oneself at rank k, the combinatorial space partitions into P(k) = (k-1)!(N-k)!, formalizing that interpersonal evaluations are projected upward or downward from an initial self-anchor:

$$P(k) = (k-1)! (N-k)!$$

Formula 5: Combinatorial Anchoring P(k)

3. Relational Dynamics: Field Regimes, Friction & Viability

3.1. Relational Friction Coefficient (Φ) and Structural Cohesion (ICE)

Collective internal resistance and misalignment are synthesized in the Relational Friction Coefficient (Φ) and its structural complement, the Structural Cohesion Index (ICE):

Φ = √(0,5 · ITA² + 0,5 · IDA²_grupo) ⇒ ICE = 1 − Φ ∈ [0, 1]

Formula 4: Relational Friction and Structural Cohesion ICE

where moderate values of Φ reflect adaptive vitality, while extreme values indicate structural strain or polarization risks.

3.2. Psychologization of Field Regimes & Groupthink Prevention

Group dynamics should not be framed as inanimate physics: collective states reflect psychosocial regimes of interpersonal articulation:

- **Regime of Structural Rigidity and Inertia:** Characterized by frozen hierarchies, static roles, and reduced communication permeability.
- **Regime of Fluidity and Adaptive Cohesion:** Characterized by open relational homeostasis, where preferences and constructive feedback circulate flexibly.
- **Regime of Dispersion and Anomie:** Characterized by low coupling density, disengagement, and weak collective grounding.
- **Regime of Effervescence and Collective Fusion:** Characterized by shared emotional activation and intense interactive mobilization.

Within this framework, NEXORD questions the assumption that optimal functioning equals absolute cohesion (ICE = 1.0, Φ = 0). A collective devoid of friction represents a vulnerable system prone to Groupthink (Janis, 1972), where consensus pressures suppress divergent perspectives. Moderate friction and transactional gradients supply the necessary impetus for organizational learning and adaptive plasticity.

3.3. The Living Puzzle: Unjamming vs. Dignified Ecosystemic Relocation

Unlike models addressing friction through punitive exclusion or individual blaming, NEXORD views the collective as a living puzzle. An actor experiencing relational tension is not inherently deficient: they often encounter structural role misalignments or flow asymmetries within their local network. Diagnostic intervention prioritizes non-punitive unjamming through local role recalibration, task restructuring, and communicative facilitation. When structural divergence proves persistent despite mediation, the framework acknowledges dignified ecosystemic relocation as a constructive option, safeguarding individual wellbeing and preventing chronic ostracism.

4. Tensorial Algebra, Q81 Lattice & Densities Pentagonam

4.1. The Dyadic Tetragram & Dual Notation

Each cell (i, j) integrates four atomic moments: D1 = pDA (emitted expectation), D2 = DA (emitted choice), D3 = REC (received choice), and D4 = pREC (received expectation).

Position / Dimension	Psychosocial Moment / Construct	Positive (+) [SMIb: A-Z]	Neutral (0) [SMIb: 0]	Negative (-) [SMIb: a-z]
D_1	pDA (Emitted expectation)	< (expects choice)	i (expects neutrality)	[] (expects rejection)
D_2	DA (Emitted choice)	E (choice)	i (indifference)	R (rejection)
D_3	REC (Received choice)	e (received choice)	? (received indifference)	r (received rejection)
D_4	pREC (Attributed expectation)	> (attributes choice)	! (attributes neutrality)	] (attributes rejection)

The lattice contains the 81 exhaustive relational configurations (Q81 = 3^4), highlighting mutual choice (<Ee>), mutual rejection ([Rr]), and reflexive indifference (i!i), decomposed analytically through 255 Boolean indicators (I255). This structured representation of signed relational patterns (+ / -) and dyadic balance connects fruitfully with contemporary research on signed graphs, communicability geometry, and structural frustration indices (Aref & Wilson, 2021; Cucuringu et al., 2023; Diaz-Diaz & Estrada, 2025).

FIGURE 5: CANONICAL 9×9 LATTICE OF THE 81 RELATIONAL CONFIGURATIONS (Q81)
 Canonical Tetragram Architecture SMla and Monotonic Valency Ranking (81 → 1)

SMla · CANONICAL DYADIC TETRAGRAM ARCHITECTURE [A1 A2 A3 A4]									
VALENCE / POLARITY	GIVING / SENDER PERSPECTIVE (DA)				RECEIVING / PARTNER PERSPECTIVE (RECIBE)				
Scale	A1: PERCEPTION (p_DA)		A2: REAL (DA)		A3: REAL (RECIBE)		A4: PERCEPTION (p_REC)		
Choice (+)	<		E		e		>		
Indifference (0)	i		¿		?		!		
Rejection (−)	[		R		r		]		

CANONICAL 9×9 Q81 NUMERICAL LATTICE (Cartesian Product DA × RECIBE · Valence 81 → 1)									
DA \ REC	e>	e!	e]	?>	?!	?]	r>	r!	r]
<E	81 <Ee>	80 <Ee!	78 <Ee]	75 <E?>	71 <E?!>	66 <E?]>	60 <Er>	53 <Er!	45 <Er]
iE	79 iEe>	77 iEe!	74 iEe]	70 iE?>	65 iE?!>	59 iE?]>	52 iEr>	44 iEr!	36 iEr]
[E	76 [Ee>	73 [Ee!	69 [Ee]	64 [E?>	58 [E?!>	51 [E?]>	43 [Er>	35 [Er!	28 [Er]
<¿	72 <¿e>	68 <¿e!	63 <¿e]	57 <¿?>	50 <¿?!>	42 <¿?]>	34 <¿r>	27 <¿r!	21 <¿r]
i¿	67 i¿e>	62 i¿e!	56 i¿e]	49 i¿?>	41 i¿?!>	33 i¿?]>	26 i¿r>	20 i¿r!	15 i¿r]
[¿	61 [¿e>	55 [¿e!	48 [¿e]	40 [¿?>	32 [¿?!>	25 [¿?]>	19 [¿r>	14 [¿r!	10 [¿r]
<R	54 <Re>	47 <Re!	39 <Re]	31 <R?>	24 <R?!>	18 <R?]>	13 <Rr>	9 <Rr!	6 <Rr]
iR	46 iRe>	38 iRe!	30 iRe]	23 iR?>	17 iR?!>	12 iR?]>	8 iRr>	5 iRr!	3 iRr]
[R	37 [Re>	29 [Re!	22 [Re]	16 [R?>	11 [R?!>	7 [R?]>	4 [Rr>	2 [Rr!	1 [Rr]

METHODOLOGICAL INTERPRETATION NOTE (NEXORD Rules 5, 8, and 10):

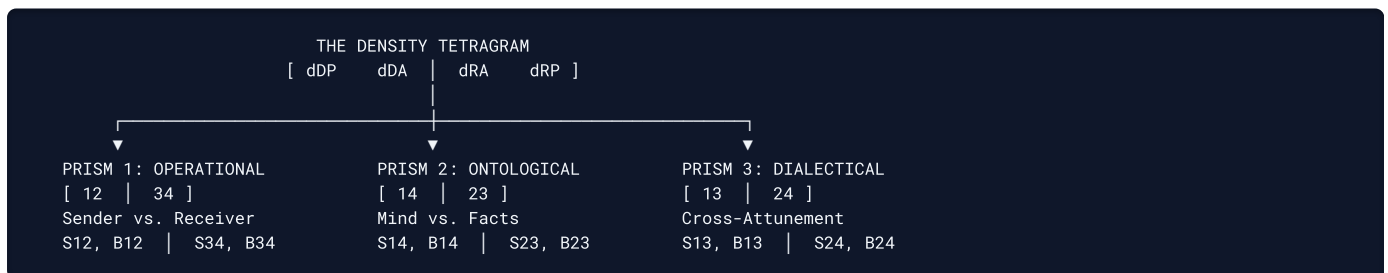
- Tetragram Architecture: Dyadic tie (i, j) = [A1: p_DA (emitted expectation), A2: DA (emitted choice), A3: RECIBE (received choice), A4: p_RECIBE (attributed expectation)].
- Q81 Lattice: Monotonic decreasing order of relational valence from 81 (<Ee>, peak mutual rejection), converging at the homeostatic centroid 41 (i¿¿, reflexive neutrality).

Table 1: Canonical 9x9 Lattice of Q81 Relational Figures

4.2. The Canonical Doctrine of the 18 Relational Densities (18D) and the 3 Dual Prisms

Micro-interactive dyadic analysis transitions from discrete rank tallies to the socio-thermodynamic continuum through the **exponential relational decay law** $W(k) = 2^{1-k}$ (1st = 1.000, 2nd = 0.500, 3rd = 0.250...). To distinguish ordinal ranks from continuous physical magnitudes, the four components of the dyadic tetragram adopt the differential prefix **d**: $\{dDP, dDA, dRA, dRP\} \equiv \{1, 2, 3, 4\}$. The continuous space operates under the universal prefix convention **S** (Signed Net Balance, $\pm\mathbb{R}$, quantifying potentials and directional flux) and **B** (Unsigned Gross Mass, $|\cdot| \geq 0$, quantifying total kinetic investment and metabolic energy).

The elemental quartet partitions into exactly **3 invariant dual prisms** ($\binom{4}{2}/2 = 3$), providing exhaustive structural refractions of dyadic reality:



1. Prism 1: Operational Axis (Sender vs. Receiver) → [12 | 34]:

Disentangles outbound expansiveness from inbound impact.

* Sender Side (Outbound): Sender Balance $S_{12} = dDP + dDA$ (SDA: net expansiveness) and Sender Mass $B_{12} = |dDP| + |dDA|$ (BDA: total emitted investment).

* Receiver Side (Inbound): Receiver Balance $S_{34} = dRA + dRP$ (SRC: induced status) and Receiver Mass $B_{34} = |dRA| + |dRP|$ (BRC: convergent relational pressure).

Diagnostics:* Uncovers power asymmetries, nurturing leaders, passive sinks, and structural ostracism.

1. Prism 2: Ontological Axis (Mind vs. Facts) → [14 | 23]:

Contrasts subjective expectations against objective interpersonal choices.

* Mental World (Crossed Expectations): Perception Balance $S_{14} = dDP + dRP$ and Perception Mass $B_{14} = |dDP| + |dRP|$.

* Factual World (Effective Choices): Action Balance $S_{23} = dDA + dRA$ and Action Mass $B_{23} = |dDA| + |dRA|$.

Diagnostics:* If $B_{14} \gg B_{23}$, diagnoses **Collective Neurosis / Group Rumination** (energy dissipated on unfounded fears and cognitive fantasies); if $B_{23} \gg B_{14}$, diagnoses **Blind Impulsivity / Reflexive Anomie** (action devoid of empathy and mutual mentalization).

1. Prism 3: Dialectical Axis (Cross-Attunement & Realism) → [13 | 24]:

Crosses sender expectation with receiver action, auditing alignment and disappointment.

* Sender Realism (Sender Expectation + Peer Action): $S_{13} = dDP + dRA$ and $B_{13} = |dDP| + |dRA|$.

* Receiver Response (Sender Action + Peer Expectation): $S_{24} = dDA + dRP$ and $B_{24} = |dDA| + |dRP|$.

Diagnostics:* Measures **latent relational friction and sociometric disillusionment** ($S_{13} \ll 0$ when $dDP > 0$).

Global system closure is established through Net Relational Potential ($SDR = SDA + SRC$), Total Kinetic Mass ($BDR = BDA + BRC$), and Positional Asymmetry ($Diff - DR = SDA - SRC$), constituting the **18 continuous relational densities (18D space)**.

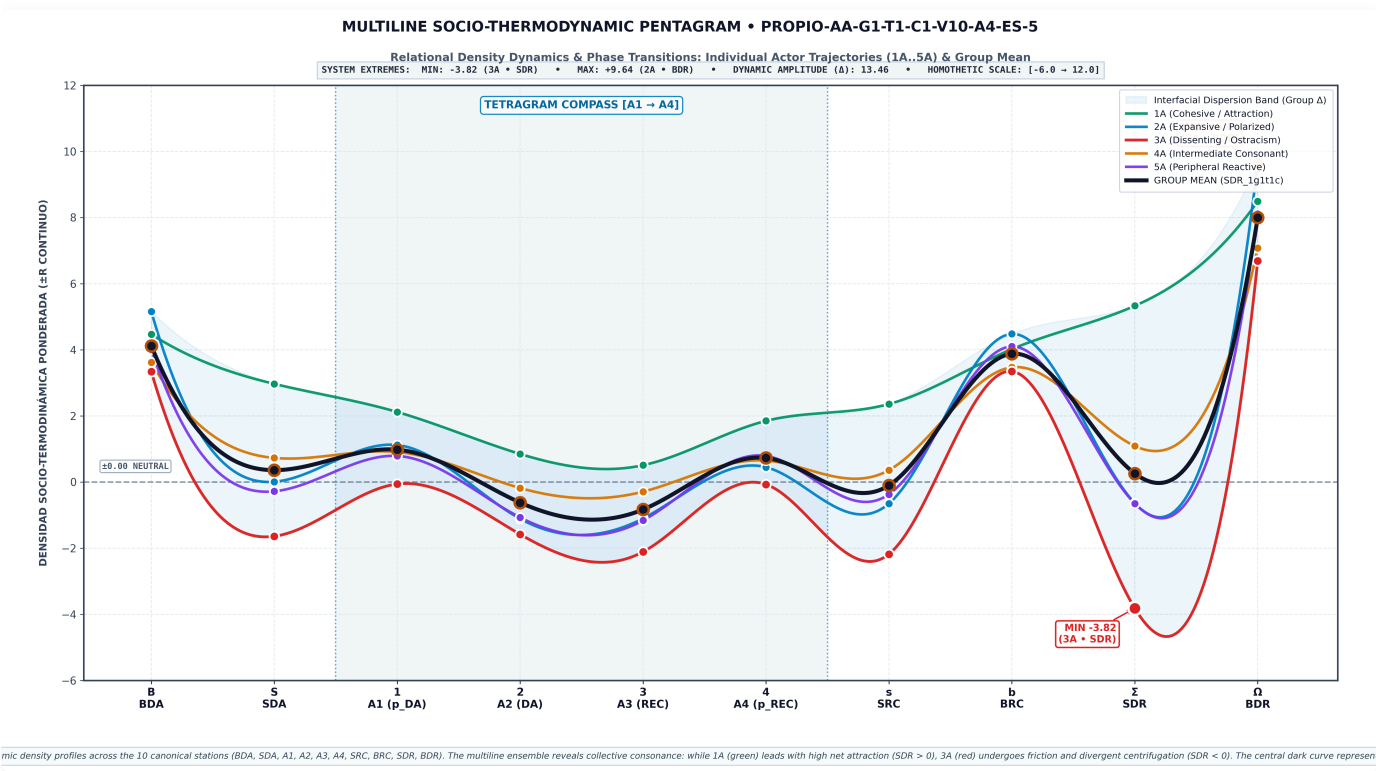


Figure 2: Socio-Thermodynamic Pentagram of Relational Densities (18D)

4.3. Continuous Quantification of the Individual Synthetic Relational Figure

Historically, aggregating an individual's collective standing through direct arithmetic averaging of discrete figures produced cancellation artifacts (blending high attractions and rejections yielded a false illusion of indifference). NEXORD resolves this by formalizing the **Individual Synthetic Relational Figure** via continuous vector decomposition:

- **Marginal Emission Density ($\bar{d}_i$):** Mean outbound relational investment toward the network: $\bar{d}_i = \frac{1}{N-1} \sum_{j \neq i} d_{ij}$ (mean expansiveness).
- **Marginal Reception Density ($\bar{d}_{-i}$):** Mean inbound social attraction converging upon the actor: $\bar{d}_{-i} = \frac{1}{N-1} \sum_{j \neq i} d_{ji}$ (mean status/in-centrality).
- **Integrated Global Density (D_i):** Canonical semi-sum of marginal densities:

$$D_i = \frac{\bar{d}_i + \bar{d}_{-i}}{2}$$

This index guarantees relational mass conservation ($\frac{1}{N} \sum_i D_i = \Delta$), scaling the physical volume, radius, and envelope of the node glyph within 3D space.

- **Relational Differential Bias (δ_i):** Normalized giving/receiving asymmetry:

$$\delta_i = \frac{\bar{d}_i - \bar{d}_{-i}}{\bar{d}_i + \bar{d}_{-i}} \in [-1, +1]$$

where $\delta_i > 0$ denotes expansive bias, $\delta_i < 0$ denotes receptive bias, and $\delta_i \approx 0$ marks symmetric reciprocity.

- **Flow Vector Angle (θ_i):** Directional angle on the (*In*, *Out*) plane: $\theta_i = \arctan\left(\frac{\bar{d}_{-i}}{\bar{d}_i}\right)$ (45° = balanced reciprocity).

4.4. The VISORD Aquarium Paradigm & Relational Hydrostatics

The **VISORD Aquarium** paradigm translates the static 2D sociogram into a continuous 3D fluid volume. Actors float and interact according to socio-thermodynamic laws:

- **X Axis:** Emission / Expansiveness (*SDA*).
- **Z Axis:** Reception / Status (*SRC*).
- **Y Axis:** Hydrostatic Depth / Buoyancy ($Z_i = D_i \times \text{scale}$).
- **Three Hydrostatic Strata:**

1. **Open Sky (Atmosphere):** Upper stratum for hyper-investing nodes exceeding water level ($Z_i > Z_{\text{sea}}$).

2. **Water Surface (Z_{sea}):** Homeostatic group equilibrium plane.

3. **Seabed / Benthos (Abyssal):** Basin of deficit buoyancy and severe ostracism ($Z_i \rightarrow 0$ or $BDF/BDC < 0.5$).

- **The Canonical VISORD Laboratories:** The platform deploys four specialized exploration suites: [VISORD_demo](#) (phase transitions and 3D/4D dynamics), [VISORD_cultural](#) (literary masterpieces with zero reactivity), [VISORD_radiológico](#) (Grassmann topological screenings), and [VISORD_dialéctico](#) (micro-interactions, social dilemmas, and IPA-SMIb Moviola), unified within the continuous volumetric hydrostatics of [VISORD_acuario](#).

5. Graphical Solutions, GxTyCz Multidimensional Design Space, and Invariant Geometry

5.1. Graphical Solutions and Parallel Dual Planes (AFC | PCA)

NEXORD deploys versatile graphical solutions to represent the continuous relational field with clarity and without information overload:

- **Three-Dimensional Continuous Sociograms:** Spatial rendering of attraction basins (neon green, <Ee>), opposition regions (orange), and friction zones (magenta, [Rr]), equipped with longitudinal temporal controls.
- **Dual Factorial Space (AFC | PCA):**

* **Correspondence Analysis (AFC):** Operates on relative contingency matrices via Chi-square (χ^2) metrics, ideal for categorical frequency profiles and sender/receiver polarization.

* **Principal Component Analysis (PCA):** Operates on continuous density metrics and the inverse rank law ($W = \pm 1/r$), preserving inertial mass and avoiding discrete ties.

* **Commutable Dual Planes:** Disentangles observations into the Senders/Givers plane (who selects), the Receivers plane (who is selected), or the unified **Factorial Biplot (F1 $\times$ F2)**. Factor 1 (horizontal) summarizes Affective Polarization (<Ee> vs. [Rr]), whereas Factor 2 (vertical) projects Expansiveness and Power Visibility (Absolute Density BDR).

- **1D Slender Vertical Axes:** Vertical projections ranking nodes by structural weight, flanked by external diagnostic tables for absolute contribution (CTA), representation quality (COR), and cumulative variance (CTR).
- **Intergroup Structurograms:** Relational support polygons supporting comparative morphological diagnostics across teams.

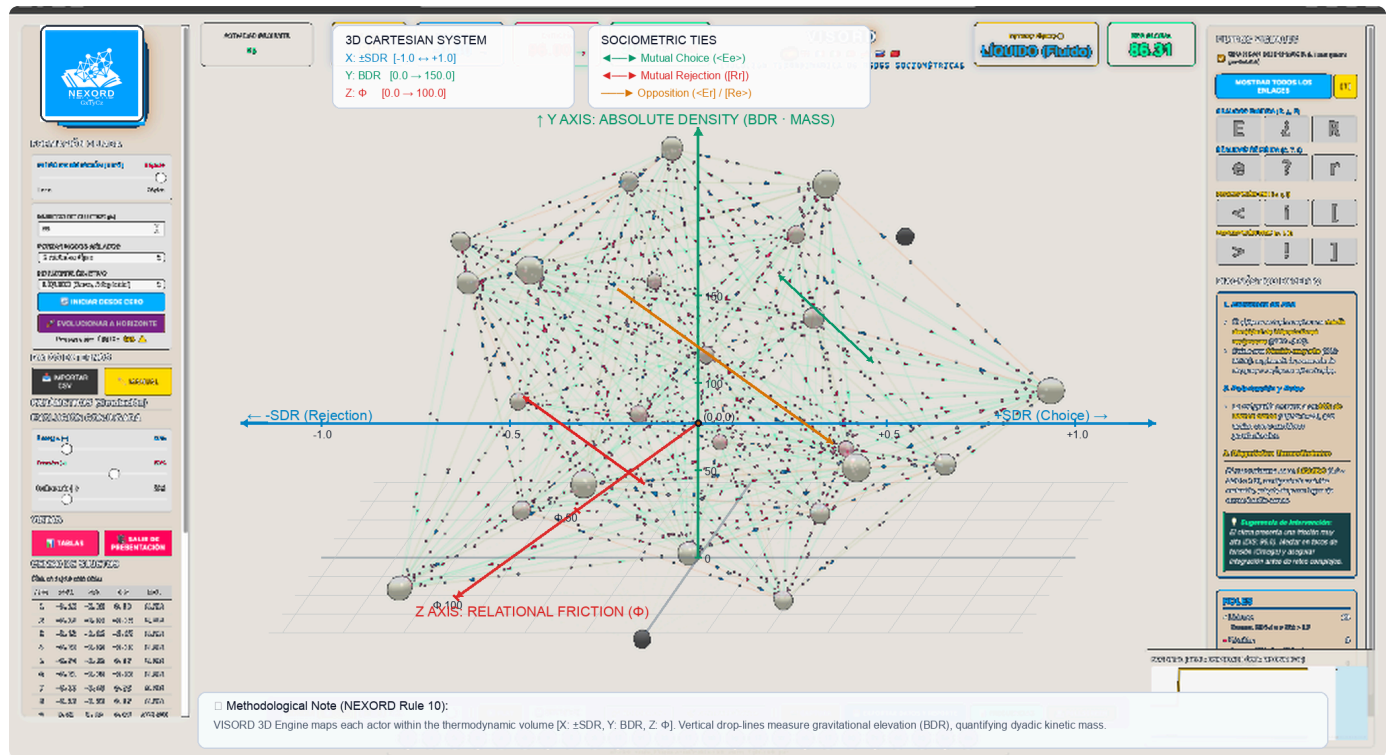


Figure 3: Graphical Solutions for 3D Relational Exploration

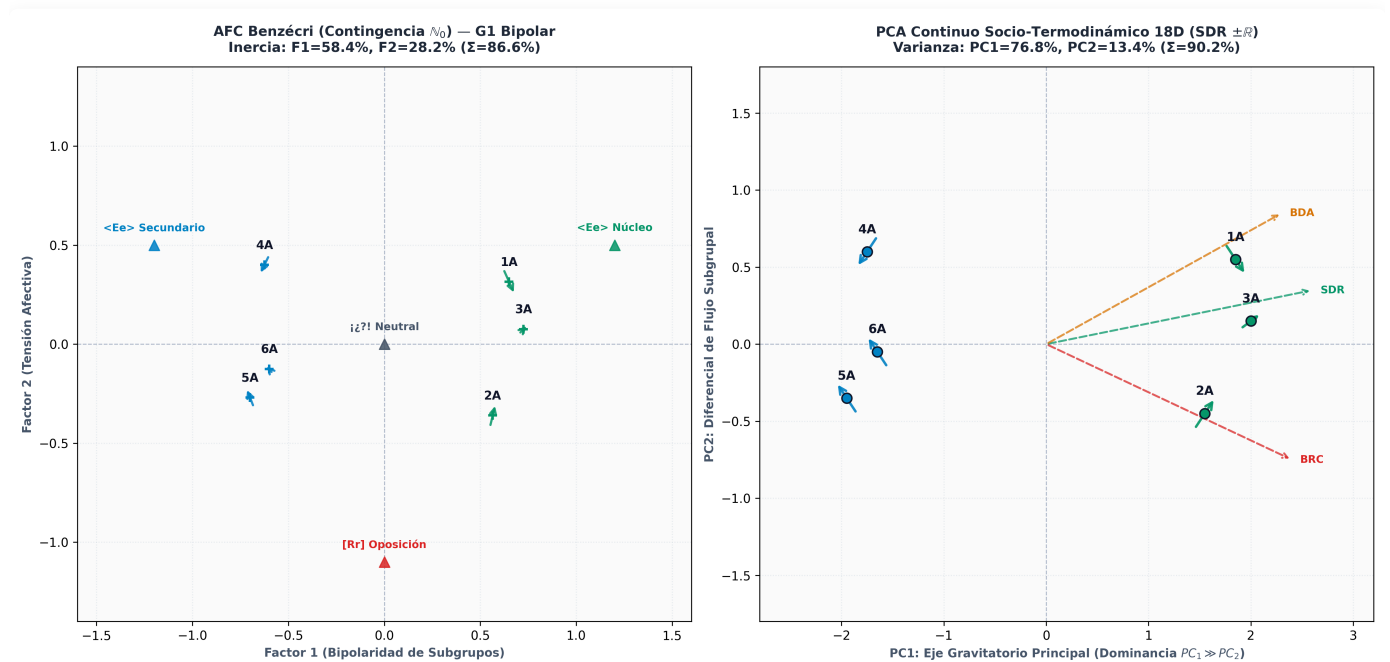


Figure 4: MESO Factorial Biplot ($F_1 \times F_2$) in Parallel Dual Space (Benzécri AFC + 10D Socio-Thermodynamic PCA). Left Panel: AFC on discrete contingency N_0 and Q81 invariant lattice. Right Panel: Continuous 10D PCA ($\pm R$) with the Densities Pentagon, positive vectors in +X and dashed projective ray extensions to -X passing through the origin (0,0). Both spaces project actors as dyadic vectorial dipoles [DA $\rightarrow$ RC] with central barycenters (+), revealing the attraction basin (2 & 1), the homeostatic bisectrix (3 & 4), and the ostracism/entropic dissipation basin (6 & 5).

Plano Factorial Comparativo Intergrup (AFC / PCA) — SIMUL-AA-G2-T1-C1-V10-A16-ES-6

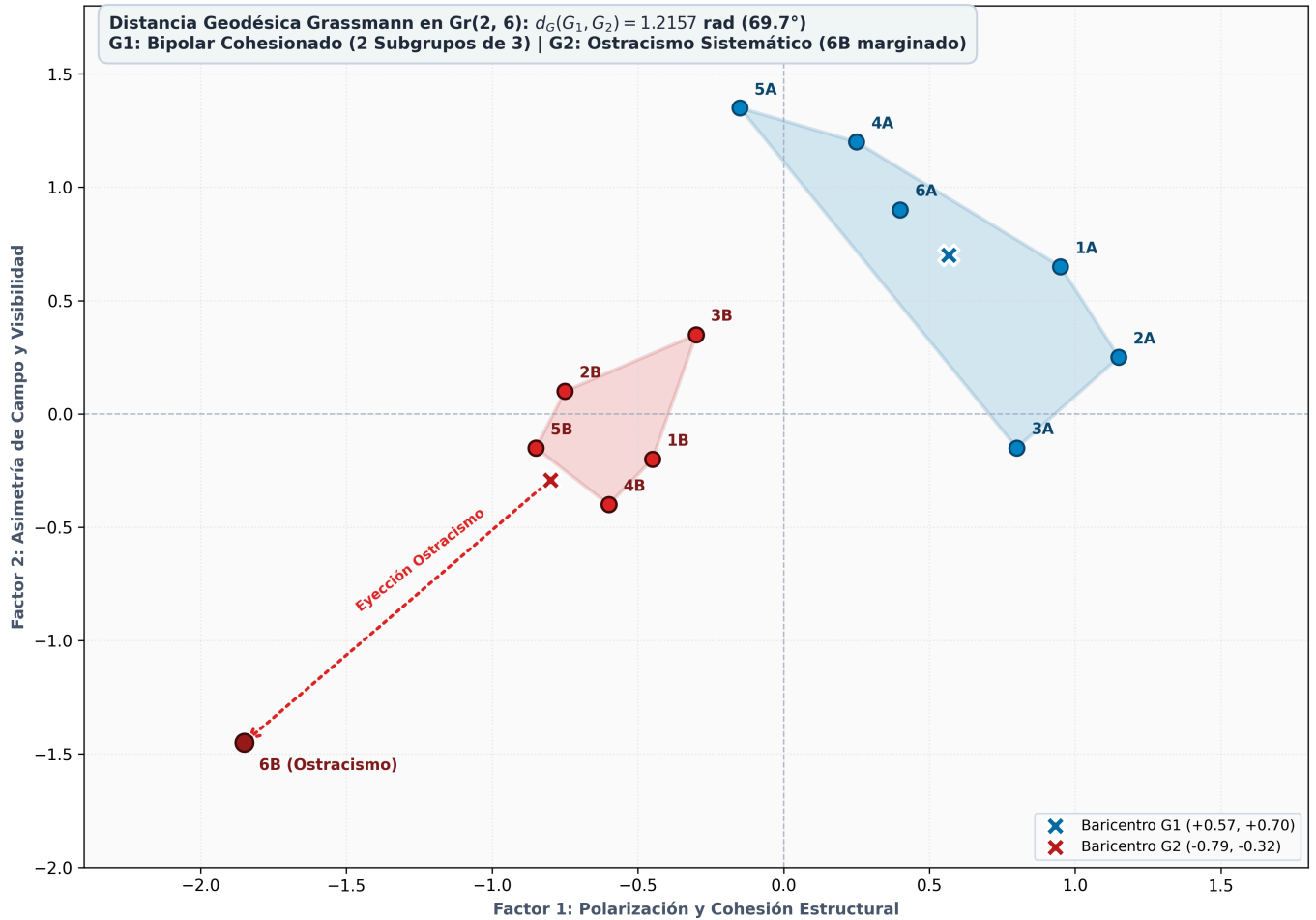


Figure 5: Intergroup Comparative Factorial Plane (AFC / PCA). Polarization force vectors and polygonal relational support structurograms between two independent collectives.

5.1.1. Paradigm Shift: From Categorical Contingency (Moreno / Benzécri) to the Socio-Thermodynamic Continuum (10D PCA)

The parallel articulation of Correspondence Analysis (AFC) and 10D Socio-Thermodynamic Principal Component Analysis (10D PCA) analytically formalizes the profound paradigm shift that Computational Ordinal Sociometry introduces beyond Moreno's classical sociometry (1934) and Benzécri's contingency approach (1973):

- 1. **Ontological Metric Nature: Categorical Discretization ($\mathbb{N}_0$) vs. Continuous Energy Continuum ($\pm\mathbb{R}$):**

Classical Sociometry (Moreno, 1934; Benzécri, 1973):* Operates on contingency tables constructed from discrete frequency tallies ($\mathbb{N}_0$) constrained by fixed quotas (typically 3 choices and 3 rejections). The Chi-square metric (χ^2) divides observed counts by the square root of marginal mass products ($\sqrt{r_i \cdot c_j}$), artificially inflating peripheral low frequencies and dispersing factor inertia across multiple orthogonal dimensions ($F_1 \approx 50.7\%$, $F_2 \approx 37.1\%$). Most critically, AFC reifies mutual rejection [Rr] as a substantive, autonomous qualitative pole symmetric to choice.

NEXORD Socio-Thermodynamics (2026):* The continuous exponential rank decay operator $W(k) = 2^{1-k}$ (1st = 1.000, 2nd = 0.500, 3rd = 0.250...) applied over tie-free forced drag-and-drop administration transforms ordinal rankings into continuous signed densities ($\pm\mathbb{R}$). Within 10D PCA space (orthogonally decomposed via SVD on the centered Pentagon matrix), all 10 canonical density variables ($SDR, BDR, SDA, SRC, BDA, BRC, D_1..D_4$) converge into the positive half-plane ($+X$). Mutual rejection [Rr], conflict, and exclusion are no longer an autonomous qualitative pole, but an **energy deficit, gravitational void, and loss of relational mass**, mapped onto the negative projective ray extensions ($-X$) passing through the barycentric origin (0, 0).

- 2. **From Static Scalar Node to Dynamic Dyadic Dipole ($DA \rightarrow RC$):**

* In classical sociograms, the individual is depicted as a static scalar dot devoid of mass or physical work, or as a passive centroid in factorial space.

* In NEXORD, the relational actor is ontologically modeled as a **kinematic vectorial dipole in continuous tension**: their active outbound investment (DA/SDA) and inbound collective reception (RC/SRC) are projected simultaneously. The directed vector arrow $DA \rightarrow RC$ quantifies socio-thermodynamic work and positional asymmetry:

$$\Delta_i = RC_i - DA_i$$

* The actor's intrinsic barycenter (+) sits with mathematical exactitude at the vector midpoint $[(DA + RC)/2]$, preserving individual systemic identity without static collapse.

- **3. The Three Topological Field Basins:**

Gravitational Attraction Basin (Leaders 2 & 1):* Upward vectors with strongly positive balances ($\Delta > 0$). The group invests massive gravitational mass in them without requiring sender hyper-effort ($RC \gg DA$). They function as gravitational potential wells that structure, polarize, and hold the collective network together at zero entropic cost.

Homeostatic Bisectrix & Critical Freezing Threshold (Intermediate Actors 3 & 4):* Subject 3 sits precisely on the homeostatic bisectrix ($RC = DA, \Delta \approx 0$), acting as a thermal damper and balancing mediator that buffers systemic fluctuations. Subject 4 marks the critical threshold of relational freezing: receiving zero positive impact ($RC = 0$), yet sustaining the group through emitted effort ($DA > 0$), accumulating severe strain ($\Delta < 0$).

Ostracism & Entropic Dissipation Basin (Peripheral Actors 6 & 5):* Collapsed, downward-plunging vectors with heavily negative balances ($\Delta < 0$). Positioned entirely within the negative projective half-plane ($-X$), diametrically opposite to all density vectors ($SDR, SRC, BDR \dots$). They act as entropic dissipation sinks absorbing collective frictions and tensions, exposing structural exclusion invisible to classical sociometric tallies.

- **4. Inertial Dominance of the Gravitational Gradient ($PC_1 \gg PC_2$):**

* In sharp contrast to AFC inertia dispersion, in socio-thermodynamic PCA the first principal component (PC_1) accounts for over 87.5% of total inertia (and up to 90% across longitudinal field corpora).

* This analytical result demonstrates that human collectives follow a **unified law of socio-affective gravity**: the relational field is not fragmented into disjointed axes, but organized along a primary dipolar gradient separating concentrated relational density ($+X$) from dissipative relational voids ($-X$).

- **5. The Barycentric Conservation Theorem:**

* In any closed human collective, the fundamental conservation law holds analytically and empirically:

$$\sum_{i=1}^N DA_i \equiv \sum_{j=1}^N RC_j$$

* Consequently, the collective mass centroid converges with floating-point precision ($\Delta_{\text{centroid}} = 0.000$, machine epsilon $\approx 2.22 \times 10^{-16}$) exactly at the Cartesian origin (0, 0). All relational energy received by an actor (RC) is strictly conserved from outbound peer investments (DA): no relational mass is created or destroyed; it is merely redistributed across the dyadic field tensor.

5.2. The GxTyCz Multidimensional Design Space without a Priori Constraints

The mathematical formulation of NEXORD imposes no a priori limitations or restrictions on research and diagnostic designs:

- **Groups (G):** Accommodates any number of independent collectives, organizational departments, working teams, or experimental groups.
- **Temporal Waves (T):** Enables continuous longitudinal tracking across T timepoints (T1, T2, ..., Tn), monitoring the diachronic evolution of cohesion and latent structural rifts.
- **Multiplex Criteria (C):** Models multiple functional and affective criteria simultaneously (C1 task competence, C2 social bonding, C3 crisis leadership), capturing topological shifts across evaluative lenses.
- **Contextual Variables (Z):** Seamlessly integrates demographic profile attributes (age, gender, tenure) and group activity scales (AAG; Vicente, Cornejo & Barbero, 2006).

Alongside core ordinal rankings, the system's formal architecture articulates two complementary psychometric modules:

- **Native Triadic Metaperceptions (aE and oE):** To complete the actor's socio-cognitive field without inflating administrative burden, NEXORD natively captures triangulated expectations projected onto the boundary poles of the relational constellation:

* **aE (Expectations on the Closest Peer):** Who does the sender expect their closest peer (1st choice) will prefer to choose?

* **oE (Expectations on the Most Distant Peer):** Who does the sender expect their most distant or rejected peer will prefer to choose?

This triadic attribution operationalizes socio-cognitive closure (following Heider and Festinger), assessing perceived transitivity, cognitive balance, and structural coherence within the group.

- **Modular Reference Batteries (V_{10} and A_{16}):** To ensure cross-group metric standardization within the Sociometric Data Bank (BSOC), the platform specifies by default a decadic sociodemographic battery ($V_{10,3}$, ten key attributes with three modalities) and the Group Adjective Attribution inventory ($A_{16,4}$, 16 canonical adjectives structured across four factors: Thematic, Functional, Cognitive, and Affective, utilizing 1-8 forced even scales). Nonetheless, both batteries remain modular, optional reference baselines: they can be modified, replaced, or tailored under the Principal Investigator's (PI) criteria according to specific substantive requirements.

5.3. Multigroup PCA & Geodesic Distance on Grassmann Manifolds Gr(k, N)

To compare formally independent collectives without enforcing artificial cross-group links, each group is modeled as a k-dimensional subspace in R^N. From canonical Jordan angles cos(theta_m) = sigma_m(U_1^T U_2), scale-invariant geodesic distance is calculated:

$$d_{Gr}(G_1, G_2) = \|\vec{\Theta}\|_2 = \sqrt{\sum_{i=1}^k \theta_i^2}$$

Formula 7: Geodesic Distance on Grassmann Manifolds

This metric quantitatively determines whether two collectives share structural morphology regardless of sample size differences.

6. Experimental Research: Physical Lab, Dialectical Sociometry, and Multimodal Diagnostics

6.1. Monitoring Experimental Designs in the Physical Laboratory

The platform supports the systematic monitoring of face-to-face experimental designs within physical social psychology laboratories. In controlled observation rooms, working seminars, and team behavioral laboratories, NEXORD captures interpersonal dynamics in real time, tracking the emergence of informal leadership structures, reciprocity, and network centrality during collaborative tasks.

6.2. Dialectical Sociometry and Social Dilemmas

Conceived as an **emerging experimental research track currently in active development**, Dialectical Sociometry investigates the real-time recording of continuous interaction sequences. Through the controlled delimitation of an **interaction space** and calibrated **interaction counters** (speaking turn frequencies, conversational pacing, and mutual listening balances), this modeling framework observes interactional dynamics in social dilemmas and collaborative tasks, monitoring the emergence of trust, gradual reciprocity, or free-riding under strict anonymization protocols.

6.3. Multimodal Perception: The Iceberg Model & Hydrodynamic Sonification

In all human groups, a visible surface layer (formal roles, explicit agreements, observable behavior) is sustained by a deep, submerged relational foundation (unspoken expectations, latent tensions, unvoiced affinities, and subtle rejections). The fundamental methodological objective of the platform is to **render visible and audible the dynamics and processes of group interaction**:

- **Visual Dimension:** Articulated through continuous graphical solutions and 3D sociograms.
- **Auditory Dimension (Hydrodynamic Sonification):** Translates friction gradients (Φ), net potential (SDR), and flow asymmetries into harmonic acoustic frequencies (calibrated at 432 Hz with distinct timbral textures). This auditory mapping allows researchers to listen to relational tension or harmony, providing a complementary perceptual modality for detecting dynamics submerged beneath the visible surface.

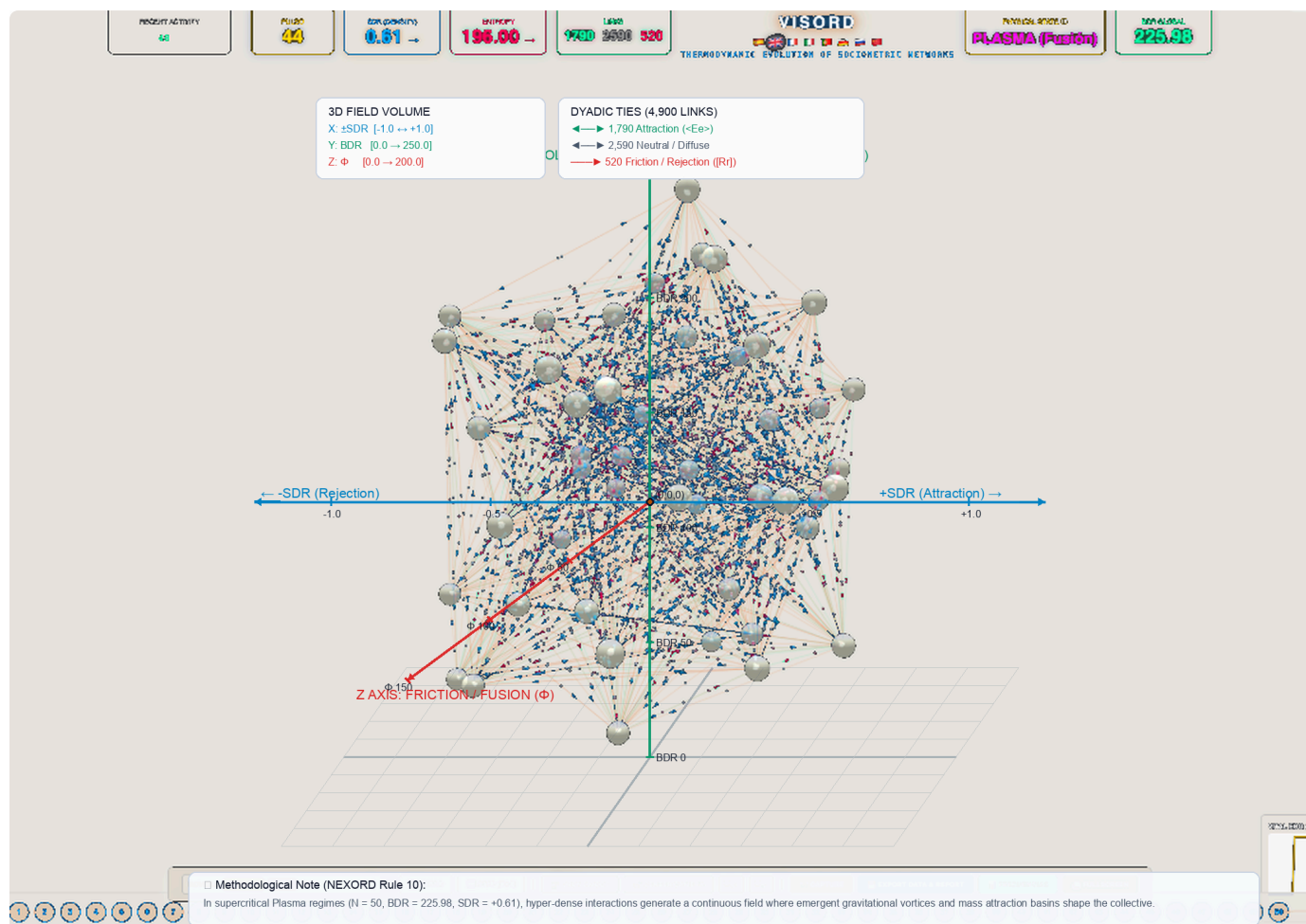


Figure 6: Continuous Field and 3D Graphical Solutions

6.4. The CULTURAL Corpus as a Zero-Reactivity Laboratory

Through the CULTURAL data category, NEXORD transcribes dramatic, cinematic, and narrative masterpieces into ordinal sociomatrices. Serving as complete and finalized relational records, literary corpora enable the forensic analysis of extreme dynamics (such as polarization or authoritarianism in works by Lorca, Sartre, or Rose) within an experimental framework with zero evaluation reactivity.

7. Institutional Governance, Ethics by Design, and Open Science

7.1. Ethics by Design & Non-Punitive Interdiction

The platform embeds ethical safeguards directly into technical algorithms (*Ethics by Design*). The system explicitly prohibits punitive deployments, negative selection, or elimination rankings. Analysis on empirical field datasets (REAL category) requires authentication and formal accountability by an accredited Principal Investigator (PI).

- **Explicit Deontological Mandate of the Principal Investigator (PI):** It is strictly prohibited to attribute strictly group socio-thermodynamic phenomena and field positions to intrinsic individual personality traits or personal merit/demerit. All sociometric diagnostics and interventions must safeguard the personal self-esteem and social reputation of participants, steering intervention toward environmental recalibration, role clarification, and collective wellbeing.

7.2. Open Science, Clélie Protocol & Institutional Custodianship (BSOC)

In accordance with Open Science and FAIR Data principles, NEXORD's mathematical, methodological, and computational formulations are publicly accessible. The complete interactive web platform, computational engines, and 3D VISORD visualizers are openly accessible at the official web portal: <https://josemancor.github.io/nex-ord-web/>. To protect participant confidentiality, empirical field datasets undergo canonical anonymization (1A1a..5A1a) and cryptographic hash sealing (SHA3-512, Clélie Protocol), under the academic custodianship of the Sociometric Data Bank (BSOC / University of Barcelona).

7.3. Participant Rights & Constructive Feedback

Participants receive individual codes to audit data processing fidelity without accessing peer records. Node re-identification is strictly restricted to the responsible PI within guidance or supportive group mediation. All feedback is constructive, non-stigmatizing, and dedicated to promoting collective health.

8. Canonical Technical Annexes

8.1. Annex I: Dual Matrices (SMIb/SMIa), Master 9x9 Lattice of Q81, and I255 Catalog

- **SMIb Matrix:** Stores the numerical tetragram $(A1 \cdot A2 \cdot A3 \cdot A4) = (pDA \cdot DA \cdot RECIBE \cdot pRECIBE)$ in each cell (i, j) .
- **SMIa Matrix:** Condenses the tetragram into the 81 dyadic configurations $(1 = [Rr], 41 = j\dot{z}?, 81 = <Ee>)$.
- **Boolean Indicator Catalog I255:** Disjunctive partitioning of dyadic space along the diagonal of perceptual consonance.

8.2. Annex II: Factorial Quality Metrics (COR, CTA, CTR), Vertical Slender Axes, and 2D(6) Projection

- **Factorial Metrics (Cornejo, 1988):**

* COR measures representation quality of node i on factor alpha.

* CTA expresses the percentage of factor inertia explained by element i.

* CTR reflects global cumulative explained variance.

- **Vertical Factorial Axes:** Slender vertical axes rank nodes by structural weight, preventing visual crowding in saturated 2D scatter plots.

Eje Factorial 1D — G1 (Subgrupos)
Equilibrio Bilateral α vs β



Eje Factorial 1D — G2 (Ostracismo)
Desplome Asimétrico de 6B

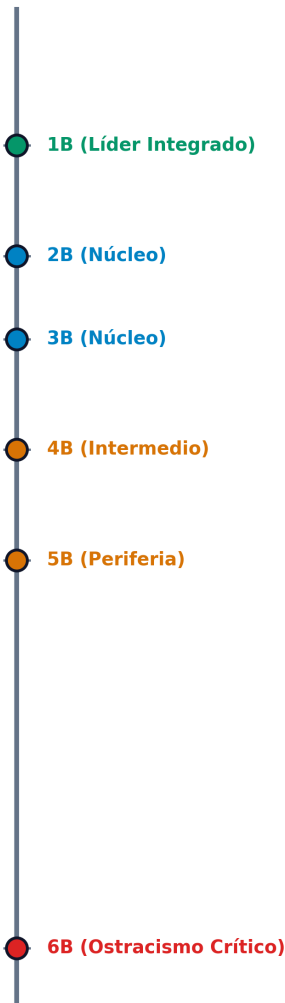


Figure Annex II: Dual 1D Factorial Vertical Axes

8.3. Annex III: Catalog of the 10 Ecosystemic Domains

NEXORD calibrates Relational Inertia (alpha) and Relational Decay (gamma) across 10 normative institutional domains:

Table Annex III: Canonical Catalog of the 10 Ecosystemic Domains

Thermodynamic Calibration Parameters (Relational Inertia α and Spatial Decay γ)

No.	Sociometric Domain	α (Inertia)	γ (Decay)	Psychosocial Focus & Operational Criteria
1	Educational & Training	0.50	0.80	Collaborative learning and inclusion (Task/Study, Play/Recess)
2	Family & Coexistence	0.70	0.90	Attachment and systemic roles (Emotional Support, Conflict Management)
3	Sports & Performance	0.40	0.70	Tactics and cohesion (Passing/Strategy, Competitive Trust)
4	Social & Community	0.60	0.85	Solidarity and social capital (Neighborly Support, Community Leadership)
5	Critical Missions & Emergency	0.30	0.60	Resilience under extreme stress (Tactical Command, Operational Support)
6	Therapeutic & Mutual Support	0.80	0.95	Alliance and emotional holding (Empathy/Listening, Self-Disclosure)
7	Work & Organizational	0.45	0.75	Efficacy and innovation (Project Collaboration, Professional Trust)
8	Institutional & Governance	0.55	0.80	Negotiation and consensus (Political Alliance, Mediation)
9	Military & Tactical	0.20	0.50	Discipline and operational brotherhood (Combat Leadership, Vital Trust)
10	Correctional & Eco-Sociometry	0.65	0.90	Coexistence and rehabilitation (Respect/Security, Support Network)

Explanatory Interpretation Note (Parameter Calibration):

1. Relational Inertia (α): Measures openness and retention of affective memory in group bonds. Higher values (e.g. Therapeutic 0.80) reflect strong affective holding.
2. Relational Decay Exponent (γ): Regulates power-law spatial attenuation of interpersonal impact. Lower values (e.g. Military 0.50) signify long-range tactical reach.
3. Multiplex Functional Alignment: Each domain pairs dual Task (instrumental) and Socio-affective (expressive) criteria ensuring balanced sociometric coverage.
4. Ecosystemic Adaptation: Field practitioners select domain parameters to calibrate thermodynamic potential and entropy calculation to institutional settings.

Table Annex III: Catalog of 10 Ecosystemic Domains

8.4. Annex IV: Sender - Receiver Rank Interdistances and Isometry Diagonal

Euclidean distances for sender emission (d_{DA}) and receiver reception (d_{RC}) preserving strict sender ordering:

- Isometry Diagonal ($d_{DA} = d_{RC}$):** Evaluates relational symmetry and verifies the Relational Asymmetry Principle (reception dispersion systematically exceeds emission dispersion).

Matriz de Interdistancias Euclídeas D_{Euc} — G1 (Subgrupos 3 + 3)

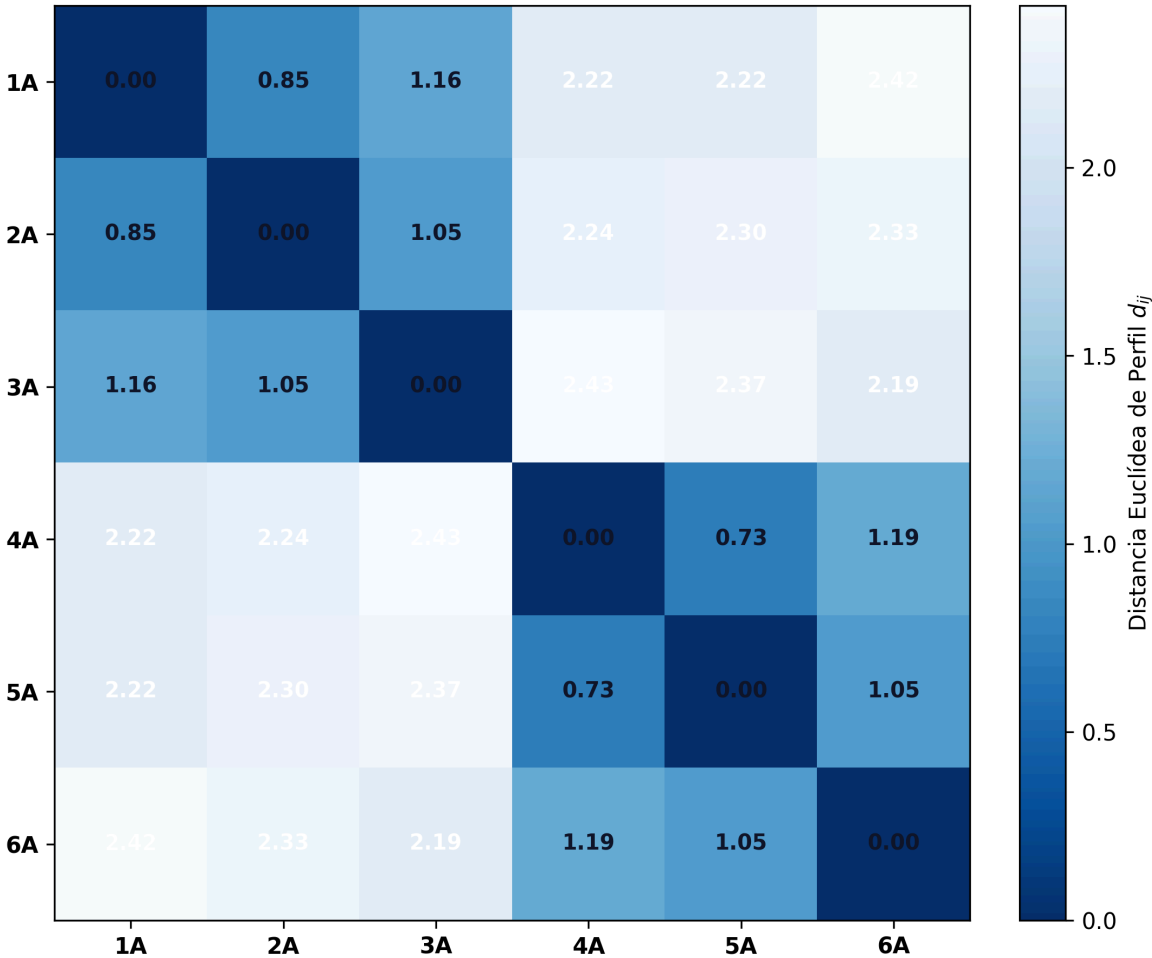


Figure Annex IV: Dispersion Plot of Euclidean Interdistances

Matriz de Interdistancias Euclídeas D_{Euc} — G1 (Subgrupos 3 + 3)

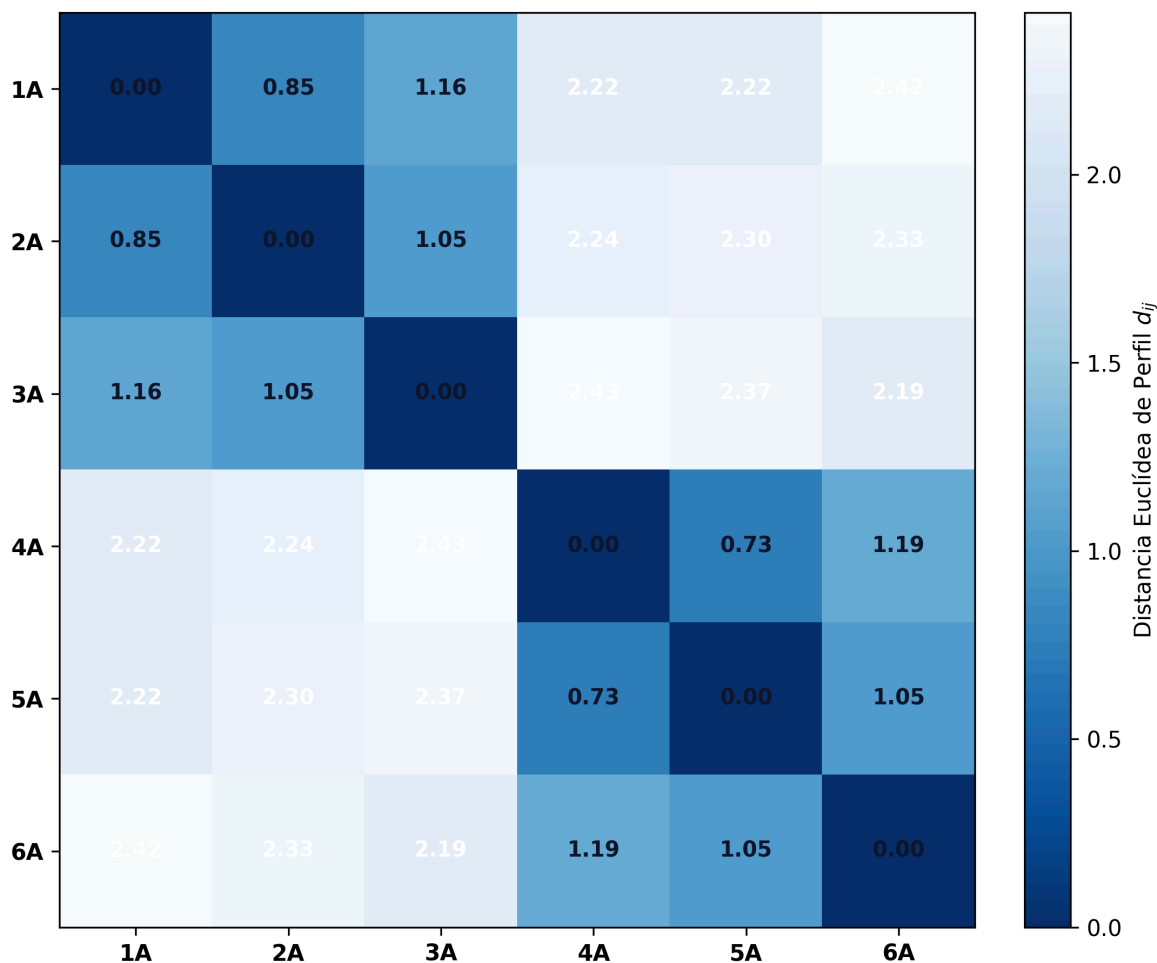


Table Annex IV: Euclidean Interdistances Table

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